

CLEANING SERVICE BOOKING MANAGEMENT SYSTEM FOR KLEENERS HUB

PROJECT PAPER

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ABSTRACT

Kleeners Hub is a desktop-based system for managing cleaning service records and bookings. It was designed to help the staff keep client and cleaner details, service information, and booking schedules in one place. The system is intended to make records easier to find and update rather than using separate lists or papers.

The system is built with Java Swing in NetBeans and uses MySQL to store its records through a database connection. After logging in, an authorized user can work with the tools permitted for their role. The system includes client and service management, booking scheduling, cleaner management, and booking records for administrators, and basic settings. Booking information includes the selected client and service, assigned cleaner when applicable, date, time, status, and service-related charges.

The project was developed in small steps so that the team could check each part and correct problems as they are found. This paper explains the reason for the system, its objectives, scope, requirements, tools, development plan, possible risks, schedule, and estimated expenses. The team should update the testing results and project dates using the actual results from the working system before final submission.

I. PROJECT BACKGROUND

1.1 Client / Organization Background

Kleeners Hub is the name used for the cleaning service agency in this project. The system is intended for staff who need to keep track of clients, cleaners, services, and bookings. Administrators manage the records and has access to the cleaner list and booking records, while staff access is limited to the functions assigned to their role.

The project is a desktop application where the Java application and MySQL database are set up. It is not a public website where customers create their own bookings; Instead, only authorized users can access the system to record and manage the cleaning service process.

1.2 Current Process

The booking process represented in the system starts when staff records or selects a client, chooses a cleaning service, and enters the booking date and time. To save a booking with its status and charges, a cleaner must be assigned; Staff need to check booking details for edits when a client changes an appointment or asks about its status.

Keeping clients, cleaner, service, and booking details in separate papers or files can make it harder to find the latest information. A central database can help the authorized users view and update related records from the application.

1.3 Identified Problems

- Client details may be difficult to find or update when they are stored in separate records.
- Staff may spend extra time checking existing bookings before creating a new schedule.
- A cleaner may be selected for two active bookings at the same date and time if the schedule is not checked carefully.
- Booking status may be unclear when changes are not recorded in one place.
- Finding bookings for a certain date or status may take longer without search or filter options.
- Service prices and booking charges need to be recorded consistently so the total amount can be checked.

2. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of this project is to develop a simple desktop system that helps Kleeners Hub organize clients, cleaners, services, and booking information. It is meant to make common record-management tasks easier and help authorized users check booking details without searching through separate records.

2.2 Main Project Objective

To develop a Java desktop application connected to a MySQL database that supports the management of cleaning service clients, cleaners, services, and bookings for Kleeners Hub.

2.3 Specific Objectives

- To allow an authorized user to log in to the system.
- To add, update/edit, delete/archive, archive list and search client information.
- To add, update/edit, delete/archive, archive list and search cleaner information.
- To add, update/edit, delete/archive, archive list and search cleaning service information.
- To create and update cleaning service bookings.
- To assign clients, services, and cleaners to a booking.
- To use booking statuses such as Pending, Confirmed, Ongoing, Completed, and Cancelled.
- To check for a cleaner's existing active booking at the same date and time before assigning that cleaner, where this check is implemented and tested.
- To view booking records and filter them by date and status. To display useful dashboard counts from the saved records.
- To provide basic settings for the system and administrator account, based on the functions available in the current version.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

The main functions planned for the current version are summarized below. Access depends on the user's role, and the final paper should be checked against the functions that work in the final build.

User / Area	Included Functions
Administrator	Log in; manage clients, cleaners, and services; schedule bookings; view booking records; use available filters; and access settings.
Staff	Log in; manage client and service records and use scheduling functions permitted by the system. Cleaners and Booking Records are restricted to administrators.
Client Management	Store client name, contact number, and address; support record maintenance and searching where implemented.

Cleaner Management	Store cleaner name, contact number, and position; available to administrators.
Service Management	Store service name, price per square meter, description, and availability/archive status.
Scheduling	Record the client, service, optional cleaner assignment, area size, date, time, property type, payment method, status, and related charges.
Booking Records	Review saved bookings and use available date/status filters; access is restricted to administrators.
Dashboard / Settings	Show selected counts and provide the settings available in the current build.

3.2 Project Limitations

- The first version is a desktop-based system and is not an online booking website.
- The system does not include online payment gateway integration.
- The system does not include payroll or advanced accounting functions.
- The system does not provide a mobile application.
- The current cleaner conflict check is based on the same date and exact booking time and does not calculate service duration.
- The system depends on the available computer, Java environment, MySQL server, and local database setup.
- Advanced reports and other future features are outside the current project scope.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project is technically possible because it uses tools that are available for student development. Java and Java Swing are used for the desktop interface, NetBeans is used to develop the application, and MySQL stores the client, cleaner, service, and booking records. JDBC is used for communication between Java and the database. The system still depends on correct database setup and a working connection.

4.2 Operational Feasibility

The system is designed around the usual tasks involved in arranging cleaning services: maintaining client and cleaner information, listing services and prices, and recording bookings. Separate screens are used for the main tasks. Users should be shown how to log in and use the functions allowed by their role before they use the system.

4.3 Economic Feasibility

The project can be developed using free or educational software. The main possible expenses are related to transportation, printing, backup storage, hardware needs, and other project requirements. Since the system is a student project, the development work itself is not included as a paid labor cost.

4.4 Project Charter Summary

Charter Item	Project Entry
Project Title	Cleaning Service Booking Management System for Kleeners Hub
Project Purpose	Improve the recording and management of cleaning service clients, cleaners, services, and bookings.
Primary User	Authorized administrator or staff user.
Major Deliverable	Working Java desktop management system with MySQL database and project documentation.
Main Constraint	Limited academic development time, available hardware, and student-level resources.
Project Success	Core management and booking functions work correctly and records can be stored and retrieved properly.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
Owner / Administrator	Provides project direction and manages important system information.

Authorized Staff User	Uses the system to manage clients, cleaners, services, and bookings.
Development Team	Gathers requirements, designs, develops, tests, and documents the system.
Clients	Provide information needed for cleaning service bookings.

5.2 Requirements Gathering Methods

The requirements listed in this paper are based on the functions planned for Kleeners Hub and the current application structure. If the team conducted interviews or observed an actual cleaning service process, the date and main findings should be added here. The methods below are possible ways to confirm requirements; they should only be described as completed if the team actually performed them.

- Interview with an intended user, if conducted, to learn what information is needed for client and booking records.
- Observation, if conducted, to understand how cleaning service requests and schedules are handled.
- Review of existing forms or records, if available, to identify the information that must be stored.
- Follow-up discussion with the instructor or intended user to confirm unclear requirements.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow an active authorized user to log in.
FR-02	The system shall allow authorized users to maintain client name, contact number, and address.
FR-03	The system shall allow administrators to maintain cleaner name, contact number, and position.
FR-04	The system shall allow authorized users to view and search service records.
FR-05	The system shall allow authorized users to add and update service name, price, and description.

FR-06	The system shall allow an authorized user to create a booking for a selected client and service.
FR-07	The system shall store booking date, booking time, status, property type, payment method, and related charges.
FR-08	The system shall allow a cleaner to be assigned to a booking when a cleaner is selected.
FR-09	The system shall allow users with permission to update booking status.
FR-10	The system shall allow administrators to view booking records.
FR-11	The system shall support the available booking search or date/status filtering options.
FR-12	The system shall distinguish archived records from active records where archive functions are used.
FR-13	The system shall show dashboard information based on records stored in the database.

5.4 Non-Functional Requirements

Category	**Requirement**
Usability	The interface should be simple enough for an authorized staff user to understand after basic training.
Performance	Common actions such as loading records, searching, and saving data should respond within a reasonable time on the target computer.
Security	Management functions should be available only after successful login.
Reliability	Saved records should remain available after restarting the application as long as the database is running and the data is not deleted.
Maintainability	Client, cleaner, service, and booking information should be updateable through the application without changing the source code.

5.5 Sample User Stories and Acceptance Criteria

User Story	Acceptance Criteria
As an administrator, I want to manage client records so that I can keep client information organized.	Client records can be added, updated, deleted, searched, and displayed in the client table.
As an administrator, I want to manage cleaners so that I can assign available cleaners to bookings.	Cleaner records can be added, updated, deleted, searched, and selected during scheduling.
As an administrator, I want to create a booking so that a client's requested service is scheduled.	A booking can be saved with a client, service, cleaner, date, time, and status.
As an administrator, I want to filter booking records so that I can find specific bookings faster.	Booking records can be filtered using a date range and status.
As an administrator, I want the system to detect a cleaner conflict so that the same cleaner is not assigned to two active bookings at the same date and time.	The system shows a warning and does not save the conflicting booking.

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

6.1 Development Methodology Used

The team uses an iterative development approach, with Scrum-style planning to divide the work into smaller tasks. The team can build a module, test it, and fix problems before moving to another module. The actual roles, sprint dates, and completed tasks should match the team's real work rather than being treated as activities that automatically happened.

6.2 SDLC Model Used

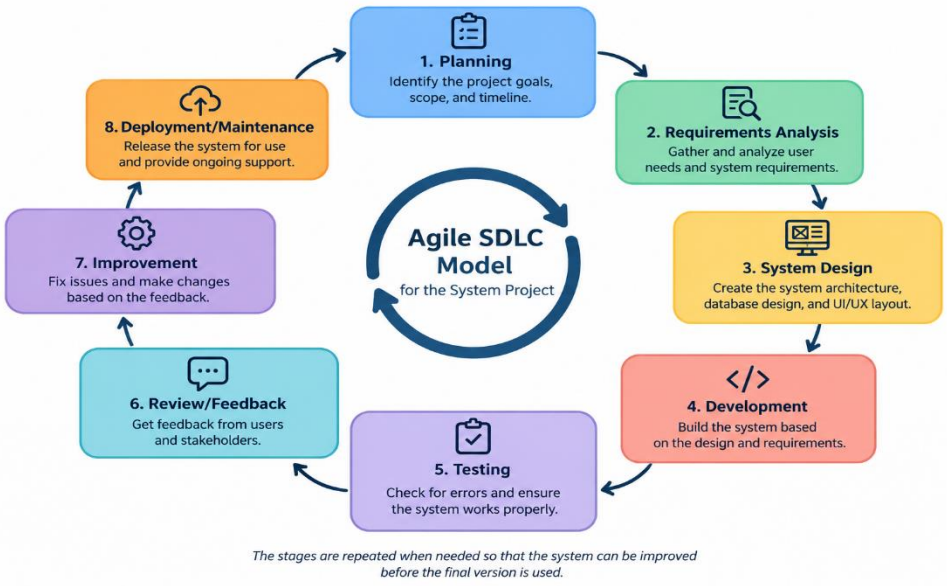
An Agile SDLC approach is used as the planned development model. The team works through planning, requirements analysis, design, development, testing, and review. These steps can repeat when the team finds an issue or receives useful feedback.

6.3 SDLC Model Figure

Figure 1. Agile SDLC Model Used for the Project

Planning → Requirements Analysis → System Design → Development → Testing → Review/Feedback → Improvement → Deployment/Maintenance

The stages are repeated when needed so that the system can be improved before the final version is used.



6.4 Software and Development Tools Used

Software / Technology	Use in the Project
Java	Used to create the desktop application and implement the system functions.
Java Swing / GUI	Used to create the forms, buttons, tables, text fields, combo boxes, and other user interface components.
NetBeans IDE	Used for developing and designing the Java desktop application.
MySQL	Used as the database management system for clients, cleaners, services, bookings, and settings.
phpMyAdmin	Used for creating, viewing, and maintaining the MySQL database during development.
JDBC	Used to connect the Java application to the MySQL database and execute SQL statements.

6.5 Reason for Selecting the Methodology and SDLC Model

An iterative approach fits this student project because the system has several connected parts. Working on one part at a time makes it easier to check whether the forms save and display the correct information. The approach also gives the team a chance to fix database and navigation problems before the final demonstration.

6.6 Expected Development Approach

- Prepare and prioritize the main system requirements.
- Divide the system into smaller features such as login, client management, cleaner management, service management, scheduling, and booking records.
- Develop and test a working set of features during each sprint.
- Check database operations and user input validation.
- Present progress and collect feedback from the instructor or project group.
- Improve the system based on accepted feedback.
- Prepare the final working version and documentation.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to Kleeners Hub
Planning	Identify the problems in managing cleaning service records, define the project objectives, scope, users, and available resources.
Requirements Analysis	Identify the information needed for clients, cleaners, services, bookings, statuses, and settings.
System Design	Prepare the GUI layout, navigation flow, database structure, and booking process.
Development	Build the login, dashboard, client, cleaner, service, scheduling, booking records, and settings modules.
Testing	Check saving, updating, deleting, searching, filtering, login, validation, and cleaner conflict checking.
Deployment	Prepare the approved desktop application and database for use on the target computer.

Maintenance	Fix discovered errors, update records or features when needed, and maintain database backups.

The team connects every SDLC stage to an actual part of the Kleeners Hub project. This helps the development process stay organized instead of only following the stages as general definitions.

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

Possible risks include changes in requirements, delays, database or source-file loss, and errors in booking information. These risks should be checked during development, especially before presenting the system.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
Requirements change during development.	Medium	High	Confirm important requirements before development and record approved changes.
Development takes longer than planned.	Medium	High	Prioritize core functions and monitor progress every week.
Project files or database are lost.	Low	High	Keep regular copies of source code and database backups.
A team member becomes unavailable.	Medium	Medium	Document completed work and share responsibilities within the team.
Target computer has limited hardware	Medium	Medium	Test the application on a computer similar to the
capability.			target device.
Users find a screen confusing.	Medium	Medium	Ask for feedback and simplify the screen or instructions when needed.

Database connection fails during use.	Medium	High	Check MySQL service, database configuration, JDBC connection, and backup setup.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The schedule below is a proposed plan based on weeks rather than confirmed calendar dates. The team should replace it with the actual dates and update the status of each activity before submission.

9.2 Project Schedule

Activity	Duration	Target Period	Main Output
Planning and Project Study	1 week	Week 1	Problem, objectives, scope
Requirements Gathering	1 week	Week 2	Confirmed system requirements
System and UI Design	2 weeks	Weeks 3–4	Interface and database design
Sprint 1 Development	2 weeks	Weeks 5–6	Login, dashboard, and basic management screens
Sprint 2 Development	2 weeks	Weeks 7–8	Client, cleaner, and service management
Sprint 3 Development	2 weeks	Weeks 9–10	Scheduling, booking records, filters, and settings
Testing and Revision	1 week	Week 11	Corrected working version
Deployment and Final Documentation	1 week	Week 12	Final system and project paper

9.3 Project Milestones

Milestone	Target
Requirements approved	End of Week 2
Initial design approved	End of Week 4
Core management functions working	End of Week 8
Scheduling and booking records working	End of Week 10

Testing completed	End of Week 11
Final presentation / documentation	End of Week 12

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The figures in the budget table are estimates for planning, not proof that the team has already spent these amounts. Replace them with the group's actual expenses or the amounts approved by the instructor.

Cost Item	**Basis**	**Estimated Cost**
Development Software	Free / educational tools	PHP 0
Database Software	Free / open-source tools	PHP 0
Transportation	Meetings, consultations, and testing	PHP 1,500
Printing / Documentation	Drafts and final copies	PHP 600
Backup Storage	USB or other backup storage	PHP 0
Hardware Adjustment	Minor peripherals or replacement needs	PHP 0
Contingency	Unexpected project expenses	PHP 900
TOTAL ESTIMATED COST		PHP 3,00

10.2 Cost Justification

The main possible expenses are transportation and printing. The development tools listed in this paper have free or educational options. The team should keep receipts or notes for expenses and adjust the table if actual costs are different.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	**Responsibility**
Product Owner	Represents the project requirements and helps prioritize the features that are important to the intended user.
Scrum Master	Helps organize Scrum activities, monitors blockers, and helps the team stay focused on the sprint goal.

Development Team	Designs, codes, tests, documents, and improves the Kleeners Hub system.

11.2 Product Backlog

Priority	Backlog Item	Expected Result
High	User Login	Authorized user can access the system.
High	Dashboard	User can see basic totals and pending booking count.
High	Client Management	Client records can be added, updated, deleted, and searched.
High	Cleaner Management	Cleaner records can be added, updated, deleted, and searched.
High	Service Management	Service records can be added, updated, deleted, and searched.
High	Scheduling	Bookings can be created and assigned to a client, service, and cleaner.
High	Booking Status	Booking status can be changed among the available statuses.
Medium	Booking Records	Saved bookings can be viewed and filtered.
Medium	Cleaner Conflict Check	The system prevents the same cleaner from being booked twice at the same date and time when the existing booking is active.
Medium	Settings	System name and administrator account information can be updated.
Low	Additional Reports	More detailed reports may be considered as a future improvement.

11.3 Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Create the basic application structure.	Login, database connection, dashboard, and navigation.
Sprint 2	Complete record management.	Client, cleaner, and service management with search functions.

Sprint 3	Complete scheduling and monitoring.	Booking creation, status updates, conflict checking, booking records, and filters.
Sprint 4	Prepare the system for final use.	Testing, bug fixing, feedback, documentation, and deployment preparation.

11.4 Scrum Events

Scrum Event	**How the Team Will Apply It**
Sprint Planning	Select the features that can realistically be completed during the sprint.
Daily Scrum / Team Check-in	Briefly discuss completed work, current tasks, and problems that need attention.
Sprint Review	Demonstrate completed functions and collect feedback from the group or instructor.
Sprint Retrospective	Discuss what worked, what caused problems, and what can be improved in the next sprint.

11.5 Simple Scrum Progress Tracking

The team can track tasks using To Do, In Progress, and Done. A task should only be marked Done after the related feature has been checked. The team should record who handled each task and the actual date it was completed.

12. PROJECT SUMMARY

Kleeners Hub is a Java desktop application intended to keep cleaning service information in one system. It connects client, cleaner, service, and booking records through a MySQL database. Its main purpose is to help authorized users record appointments, review booking details, and keep service information organized.

The current project is focused on basic management and scheduling. It is not an online customer booking website and does not include online payment, payroll, advanced accounting, or a mobile application. The paper should only claim that a function is complete after the team has tested it in the application.

The system uses Java, Java Swing, NetBeans, JDBC, and MySQL. The team is following an iterative development plan so the main functions can be developed and checked in parts. Database records include client contact details, cleaner details, service name and price, and booking details such as date, time, status, and charges.

Overall, the system is intended to make daily cleaning service records easier to manage. Before final submission, the team should verify each function, record actual test results, confirm the project schedule and budget, and make sure the paper describes only what the current version can do.